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A novel image encryption algorithm based on self-adaptive wave transmission

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ABSTRACT

Based on self-adaptive wave transmission, a novel image encryption algorithm is given in this paper. The advantages of our proposed approach are that it can be realized easily and is computationally simple while achieving high security level, high speed, high sensitivity and other properties simultaneously. Wave transmission encryption, as the name suggests, is a way to change the gray-level value of pixels by simulating waves' transmission. The self-adaptive encryption is carried out by using one half of image data to encrypt the other half of the image mutually. Our algorithm can encrypt image in parallel and be also applied to color image encryption. Simulation results for a gray-level and a color image have demonstrated the high performance on the sensitivity, speed, and security of the proposed algorithm. Even in the first round the change rate of the number of pixels in the cipher-image when only one pixel of the plain-image is modified (NPCR) and the unified average changing intensity (UACI) are already very high ($NPCR > 0.995$, $UACI > 0.334$), only 2 rounds encryption can satisfy the performance and security requirement. Besides, our algorithm is faster than state of the art techniques, as far as parallel encryption is concerned, the time consumption will be much less.

1. Introduction

With the ever-increasing growth of multimedia applications, security has become an important issue on communication and storage of images. Usually, encryption is one better way to ensure the high security. It is well-known that image encryption has extensively applications in Internet communication, multimedia systems, medical imaging, telemedicine, and so on.

In spatial domain, the basic ideas for image encryption can be classified into three major types: the position permutation, the value transformation, and their compounding form. There already exist several image encryption methods in spatial domain, among which the tree structure-based methods [1], chaos-based methods [2–9], 2D cellular automata-based methods [10,11] are most popular.

A tree structure-based method is proposed in [1], in which the quadtree structure is selected to be encrypted. It has an advantage of reducing the processing time for encryption and decryption. However, it is suitable only for the quadtree compression algorithm, but it is not an international standard.

So far, a variety of chaos-based algorithms [2–9] were proposed for image encryption. Because chaotic systems possess several pretty properties, such as randomness, sensitivity, simplicity, ergodicity, chaos have been used as essential components to constructing cryptosystems. The two-dimensional or the high dimensional chaotic maps which can be considered as a 2D array of pixels are naturally employed to encrypt image. By applying chaotic systems, those algorithms can achieve good performance. A chaos-based image encryption scheme typically composes the iterations of two processes: substitution and

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diffusion. The former is achieved by permuting all the pixels as a whole using a 2D chaotic map (such as cat map, baker map, tent map, magic map) [3,9]. In the diffusion process, the pixel values are altered sequentially and the change made to a particular pixel depends on the accumulated effect of all the previous pixel values. This architecture forms the basis of a number of chaos-based image ciphers proposed. There are still some chaos-based image cryptosystems with their own structure [5,7]. However, as many rounds of substitution and diffusion or iteration should be taken, the overall encryption speed is slow. Though many improvements are presented recently, the speed is still no match to our algorithm. Moreover, there still some security weaknesses in some chaos-based encryption. For example, Cahit, etc. have presented a complete break of the chaos-based image encryption algorithm proposed [14].

Cellular automata (CA) [10,11] is also introduced into encrypting images because it has the following advantages: (i) Since CA involves local interactions, it has been applied successfully to several physical systems, processes and scientific problems, especially in image processing data encryption and byte error correcting codes. (ii) The number of CA evolution rules is very large. Hence, many techniques are available for producing a sequence of CA data to encrypt and decrypt images. (iii) Recursive CA substitution only requires integer arithmetic and/or logic operations, so it simplifies the computation burden. Recently, an image encryption algorithm by using recursive cellular automata substitution is proposed to replace the pixel values in CBC-like mode in [11]. It is obvious that their algorithm neither can resist known-plaintext attack nor can be implemented in parallel.

Based on the discussion above, although many encryption methods have been proposed, each of them has its strength and limitation more or less in terms of security level, speed, and sensitivity, etc. almost none of them can achieve them all simultaneously.

Wave is a common phenomenon, the shape of which is determined by three parameters named: source point, wavelength, and amplitude (see Fig. 1). An image can be treated as a calm lake surface to let discrete "waves" transmit on it. The gray -level value of the pixels in the image is changed by adding the corresponding displacement of the wave to the original gray-level value according to the distance between the pixel and the source point, then modulus 256. Since wave is periodical, every pixel of the image can be interfered by the waves.

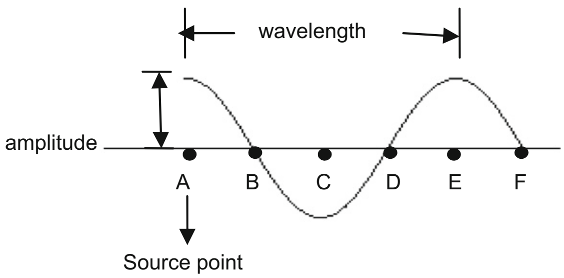

Fig. 1. The shape of a wave.

In this paper, a novel image encryption algorithm based on self-adaptive wave transmission is proposed. It can be easily implemented and is computationally simple while achieving high security level, high speed, and high sensitivity and other properties simultaneous. Moreover, as shown in Section 3, it can encrypt image in a parallel mode and be applied to encrypt color image. Simulation results for a gray-level image and a color image demonstrate the high performance on the sensitivity, speed, and security of the proposed algorithm.

This paper is organized as follows. The basic idea of self-adaptive wave transmission encryption is described in Section 2. The proposed algorithm is presented in Section 3. Simulation results are given in Section 4. The security and performance analysis are shown in Section 5. Finally, some conclusions are presented in Section 6.

2. Problem formulation

As mentioned in the above part, for chaos-based algorithm, many rounds of substitution and diffusion or iteration should be taken. For instance: the parameters recommended for Shiguo Lian's scheme [4] are $m=6$, $n=3$ while for Kwok-Wo Wong's scheme [8] $m=1$, $n=4$ where m means the number of the whole substitution–diffusion rounds and n stands for the number of permutation rounds (maybe at the same time do the value substitution) in the substitution stage. This mainly results from two facts: one is that the diffusion effect is contributed mainly or solely by the diffusion process. To meet the general needs of NPCR and UACI ($\text{NPCR} > 0.995$, $\text{UACI} > 0.334$), many rounds are needed. The other one goes that inadequate rounds cannot guarantee the security [14].

As for 2D cellular automata-based methods, on one hand, due to the born contradiction between CBC-like mode and parallel mode, if CBC-like mode has not been eliminated, parallel mode, which can doubly reduce the encryption time, cannot apply to encryption. On the other hand, to resist known-plaintext attack, diffusion process should be introduced in.

Wave is a common phenomenon, the shape of which is determined by three parameters named: source point, wavelength, and amplitude (see Fig. 1). When a wave transmits on a lake, the surface will go up and down accordingly and it is noticeable that a little change on any of the three parameters can cause a totally different surface. Similarly, an image can be treated as a calm lake surface to let wave transmit on it and the gray-level value of the pixels in the image can be changed according to the wave. Since the wave is periodical, every pixel of the image can be interfered by the wave. It can be predicted that it is computationally simple and only simple operations such as XOR operations; addition, etc. will be involved.

Before employing wave transmission, two problems should be settled; one is that due to the periodicity, one wave is not safe for encryption. The other is that wave

Fig. 2. The demonstration of one round self-adaptive encryption.

transmission is not sensitive to the plaintext. When the gray-level value of a pixel in plain-text is changed, only the gray-level value of the corresponding pixel in cipher-image will be changed, which makes it vulnerable to known-plaintext attack.

Our solution to the first problem is simulating several waves. In real-life, many waves can be transmitted on a single lake as a phenomenon called superposition of waves. So, to enhance the performance of our algorithm, we choose to simulate several waves defuse such a disadvantage. The effect of the superposition of waves equals to superposition of the effect of every wave. In our algorithm, four sin waves are chosen to be simulated. The sin wave is chosen because it is the one resembles a real wave most. The number of waves is a compromise between performance, security and computation cost. The decrease in the number of wave will not guarantee the performance and security, while the increase in the number of wave will add to the computational consumption.

An elegant way to solve the second problem is self-adaptive encryption [12], which is an excellent tool in image encryption. The basic idea of self-adaptive is to divide image into two parts equally. First, the information of one part is used to encrypt the other part, in turn; the encrypted part is used to encrypt the other one. For example, see Fig. 2. When integrate it with wave transmission, the information mentioned here in our algorithm can be used to determine the three parameters of the waves. Then a little change in plain-image will generate different parameters. As the wave is sensitive to its three parameters, so it will bring huge difference to cipher-image. This means self-adaptive wave transmission encryption possess high sensitivity.

Additionally, according to the discussion below, it can be seen that self-adaptive wave transmission encryption can be encrypt image in a parallel mode.

3. The proposed algorithm

We propose an encryption algorithm based on self-adaptive wave transmission. During encryption, an image is divided into two parts equally, both of which are encrypted by using wave transmission encryption with four waves whose amplitudes are determined by the other one.

3.1. Gray-level value transformation of a pixel

First, the gray-level value transformation of a pixel encrypted by only one wave is described. The superposition of multiple waves can be implemented by simply adding the transformation result of each wave. In this paper, λ is used to represent the wavelength and A for amplitude.

Once the three parameters are fixed, according to the shape of the discretized wave, the value transformation s of the gray-level value of the pixel can be obtained by using the following formula:

$$s = round\left(A\sin\left(\frac{2\pi d}{\lambda}\right)\right), \tag{1}$$

where d is the distance between the pixel and the source point, which can be got as follows,

$$d = \sqrt{(m\text{-}x)^2 + (n\text{-}y)^2}, \tag{2}$$

where (m, n) is the coordinates of the source point, (x,y) is the coordinate of the pixel.

Then, the new Gray-level value can be obtained easily as follows,

$$V_{out} = \mathrm{mod}(V_{original} + s, 256) \tag{3}$$

where $V_{original}$ is the original gray-level value of the pixel, V_{out} is the gray-level value after encryption, mod(x,y) returns the result of x modulus y.

From those formulae, it is can be seen that a little change on any of the three parameters can make difference on the final gray-level value for all the pixels involved.

It also can be noticed that the range of the displacement is in direct proportion to A while the rate of change of displacement is in inverse proportion to λ, which can be utilized to improve the performance of our algorithm. Due to the intrinsic features of image on high correlation, if A is not big enough and the λ is too large, let Δ_1,Δ_2 represent the two distances between a pair of two nearby pixels and the source point, respectively. Since the differences between Δ_1 and Δ_2 are very slight, the displacements of the two nearby pixels will be also very close, which will reveal the outline of the plain-image in cipher-image. On the other hand, if A is over large, it will be not convenient to represent in computer and make the computation complex.

3.2. The process of the algorithm

(1) Set i=0. Pad the W *H image by replication of the right-most column and the bottom row to make sure that the number of rows and columns in this image are both a multiple of 4. Here W and H are the width and height of the image.

(2) If i is even, partition the image into two equal parts in vertical direction, else partition the image into two equal parts in horizontal direction.

(3) Encrypt the image. Take the situation that i is even for example (when i is odd, the process can be carry out similarly).

(a) Divide the lower part into 2×2 blocks equally and get four values $p(n)$ (n=1, 2, 3, 4) by XORing all the gray-level values within a block.

Let $g(i, j)$ be the gray-level of the pixel at row i, column j in the part, then the value $p(n)$ got in block n can be obtained as follows:

$$p(n)=\sum_{((i,j)\in block(n))} g(i,j).$$

Note that the addition here is defined as the bitwise XOR operation.

(b) Generate the three parameters for four waves.

For the ith wave, the coordinates of the source point (m, n) are got as follows:

$$m=\mathrm{mod}\left(XOR(p(i),key_1(i)),\frac{W}{2}\right)n=\mathrm{mod}(XOR(p(i),key_2(i)),H),$$

where $key_1(i)$ (i=1,2,3,4), $key_2(i)$ (i=1,2,3,4) are our keys, the modulus operation are employed to keep the coordinate of the source point within the upper part.

For the ith wave, the wavelength λ are obtained as follows:

$$\lambda=\frac{XOR(p(i),key_3(i))}{2}+1,$$

where $key_3(i)$ (i=1,2,3,4) are our keys, the division are used to keep the wavelength short, the addition are used to avoid the situation that λ=0.

For the ith wave, the amplitude A is obtained as follows:

$$A=128\times(XOR(p(i),key_4(i))+1),$$

where $key_4(i)$ (i=1,2,3,4) are our keys, the multiplication are used to keep the amplitude large. The addition are used to avoid the situation that A=0.

AS there are four waves, so the total length of our key is $4\times 4\times 8$=128 bits.

(e) Encrypt the upper part by using wave transmission encryption.

(f) Go to (a), use the upper part to encrypt the lower one.

(4) Set i=i+1. Go to step (2) until the cipher image reaches our security requirements.

3.3. Decryption

The decryption procedures are typically composed of a reversed order of the encryption performed in encryption. This property also holds in our scheme.

As discussed above, when encrypting one part of the image, we keep the other part as it is and use its information to encrypt the selected part. So, in decryption, those information can also be used, together with our key, to get the value transformation s. Then, with s known, according to formulae (3), the original gray-level value of the pixels can be obtained as follows:

$$V_{original}=\mathrm{mod}(V_{out}\text{-}s,256),$$

Let $C(n)$ denotes the cipher-image after n rounds encryption, $C(0)$ stands for plain-image. Let $C_L(n)$,$C_R(n)$, $C_A(n)$,$C_B(n)$ $n=0,1,2,3\ldots$ denotes the left half part, the right half part, the upper half part and the lower half part of the cipher-image after n rounds encryption respectively. Here we explain how to use $C(n)$ and key to get $C(n-1)$, plain-image (i.e. $C(0)$) can be got by following those steps certain times.

Suppose in the nth round encryption we first encrypt the upper part using the information of the lower part, then encrypt the lower part using the information of the upper part. See Fig. 3. Then we get $C_A(n)=E(C_A(n\text{-}1),C_B(n\text{-}1),key)$ and $C_B(n)=E(C_B(n\text{-}1),C_A(n),key)$. Here E means the encryption process, in which the first parameter means the part being encrypted, the second parameter means the source of p and the last one is the key.

In decryption, as $C_A(n)$ and key are known, $C_B(n-1)$ can be got using the corresponding decryption process. That is $C_B(n\text{-}1)=D(C_B(n),C_A(n),key)$. Here D means the decryption process, in which the first parameter means the part being decrypted; the second and the last parameter are the same to E. $p(n)$ is got by XORing over all pixels in the same block of $C_A(n)$ Now with $C_B(n-1)$ known, together with key, $C_A(n-1)$ can be obtained similarly. As $C_B(n-1)$ and $C_A(n-1)$ are known, $C(n-1)$ has been obtained successfully.

3.4. Parallel image encryption

To allow parallel image encryption, the conventional CBC-like mode must be eliminated. However, this will cause a new problem, i.e. how to fulfill the diffusion requirement without such mode. Besides, there are some additional requirements for parallel image encryption [11]:

(1) Computation load balance

The total time of a parallel image encryption scheme is determined by the slowest PE, since other PEs have to wait until such PE finishes its work. Therefore a good parallel computation mode can balance the task distributed to each PE.

(2) Communication load balance

There usually exists lots of communication between PEs. For the same reason as of computation load, the communication load should be carefully balanced.

(3) Critical area management

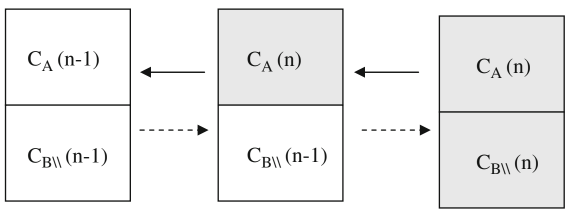

Fig. 3. The demonstration of one round decryption (the gray parts mean the parts that have been encrypted). Solid arrow: decryption direction; Dotted line: encryption direction.

When computing in a parallel mode, many PEs may read or write the same area of memory (i.e. critical area) simultaneously, which often causes unexpected execution of the program. It is thus necessary to use some parallel techniques to manage the critical area.

Since the conventional CBC-like mode is eliminated, the proposed algorithm potentially can be used flexibly to encrypt image in a parallel mode. Here is a example for encrypting image with our algorithm in a parallel mode. See Fig. 4. The image to be encrypted is divided into 8 parts equally, each of which is encrypted by a single processing element (PEs) simultaneously. First, four of them are selected to determine the three parameters for four waves, with which the other four parts are encrypted in parallel. Then, in turn, the encrypted part is used to encrypt the others in parallel. The data needed to be exchanged during encryption are only the three parameters of the four waves.

It is noticeable that our algorithm cannot only fulfill the diffusion requirement but also the other three additional requirements.

3.5. Applying the algorithm to color image

A color image comprises three components, i.e., R, G and B, which can be seen as three parts, then the process of encrypting gray-level image can be applied to the color image similarly: First, arrange the three components in a certain order. Then according to the order, use the first one to encrypt the second one, the second one for the third one, the third one for the first one, repeatedly until the cipher image satisfies our performance requirement.

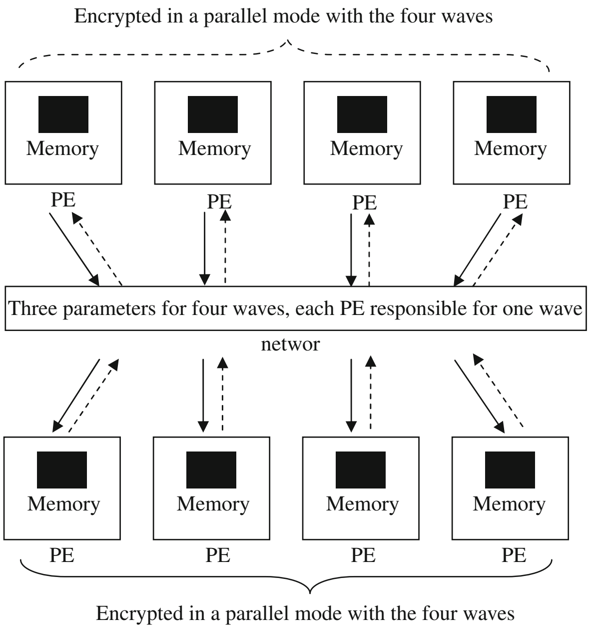

Fig. 4. One round of parallel encryption. Solid line: four of the 8 parts are selected to encrypt the others. Dotted line: in turn, the encrypted part is used to encrypt the unencrypted ones.

4. Experimental results

In this section, an example is given to illustrate the effectiveness of the proposed algorithm. In the experiment, one gray level image 'Lena' and one color image 'FuYuanAi' of size 256*256 pixels, is chosen as the plain-image. The plain, cipher images are presented in Fig. 5.

Typically, we encrypt the plain-image for 2 rounds as recommended for good performance.

4.1. Histogram

Histograms of the plain-image and the cipher-image after 2-round encryption are depicted in Fig. 6, respectively. These figures show that the cipher-image possesses the characteristic of uniform distribution in contrast to that of the plain-image.

4.2. Correlation analysis of two adjacent pixels

The correlation analysis is performed by randomly selecting 2000 pairs of two adjacent pixels in vertical, horizontal, and diagonal direction, respectively from the plain-image and the ciphered image. Then the correlation coefficient of the pixel pair is calculated and the result is listed in Table 1. Fig. 7 shows the correlation of two horizontally adjacent pixels after 2-round encryption. It is evident that the neighboring pixels of the cipher-image have little correlation.

4.3. Sensibility analysis

4.3.1. NPCR and UACI analysis

NPCR means the change rate of the number of pixels of the cipher-image when only one pixel of the plain-image is modified. The unified average changing intensity (UACI) index measures the average intensity of differences between two images. Let us assume two ciphered images C_1 and C_2 whose corresponding plain images have only one-pixel difference. The gray-level values of ciphered images C_1 and C_2 at row i, column j are labeled as $C_1\ (i,j)$ and $C_2\ (i,j)$, respectively. NPCR and UACI are defined as:

$$NPCR = \frac{\sum_{i=1}^{W}\sum_{j=1}^{H} R(i,j)}{W \times H} \times 100\%,$$

$$R(i,j) = \begin{cases} 1 & c_1(i,j) \neq c_2(i,j), \\ 0 & c_1(i,j) = c_2(i,j), \end{cases} \tag{4}$$

$$UACI = \frac{\sum_{i=1}^{W}\sum_{j=1}^{H} \frac{|c_1(i,j)\text{-}c_2(i,j)|}{255}}{W \times H} \times 100\%, \tag{5}$$

where W and H are the width and height of the image.

In our example, the selected pixel is the last pixel of the plain-image. For Lena, its value is changed from $(01101011)_2$ to $(01101010)_2$. For FuYuanAi, only the value of R plane changed from $(000001111)_2$ to $(00001110)_2$. Then the NPCR and UACI at different rounds are calculated and listed in Table 2. The data show that the performance is satisfactory even in the first rounds of encryption. The different pixels of 'Lena' cipher-images and R plane of

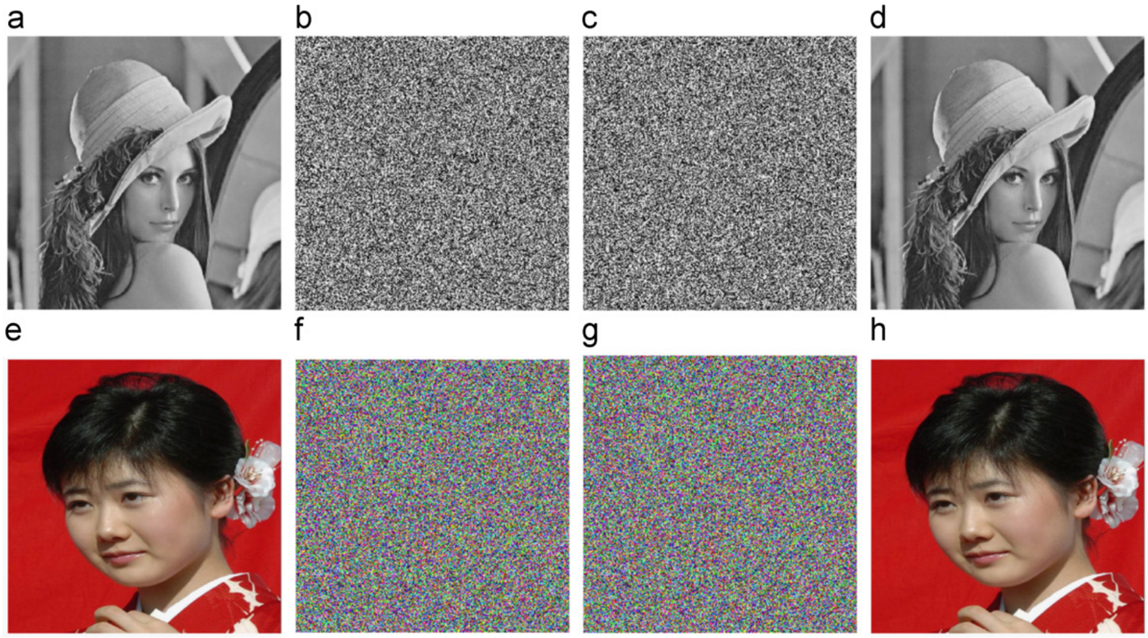

Fig. 5. The plain, cipher, decrypted images. (a) Plain-image Lena, (b) cipher-image, 1-round, (c) cipher-image, 2-round, (d) the decrypted image, (e) plain-image FuYuanAi, (f) cipher-image, 1-round, (g) cipher-image, 2-round, and (h) the decrypted image.

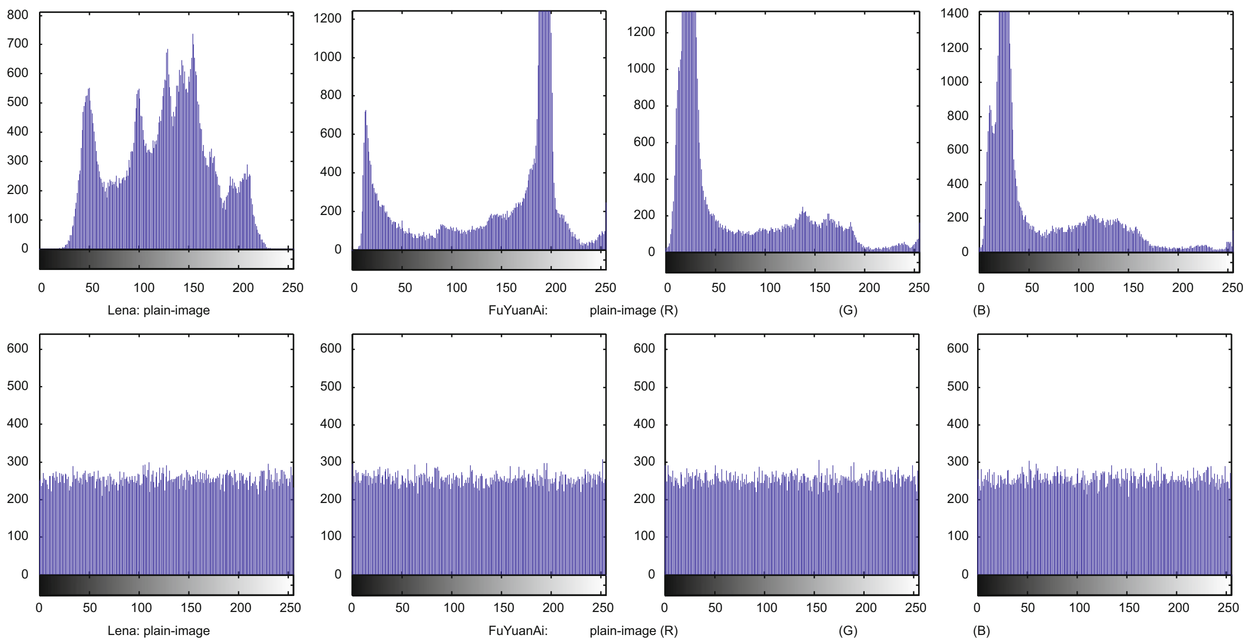

Fig. 6. Histogram of plain-image and cipher-image of the two images.

'FuYuanAi' cipher-images after 3 rounds of encryption are plotted in Fig. 8.

Table 3 compares the NPCR and UACI for the proposed scheme, Kwok-Wo Wong's scheme [8] and Shiguo Lian's scheme [4] on 'Lena'.

It can be seen that our algorithm possesses better performance on the sensitivity over the other two.

4.3.2. Key sensibility

Key sensibility means the change rate of the number of pixels of the cipher-image when only one bit of the key is modified. In our example, for key_1 (1), its value is changed from 171 to 170. For key_2 (2), its value is changed its value is changed from 95 to 94. For key_3 (3), its value is changed from 211 to 210. For key_4 (4), its value is changed from

Table 1
Correlation coefficient of two adjacent pixels in plain-image and cipher-image.

	Lena		FuYuanAi					
	Plain-image	Cipher-image	Plain-image			Cipher-image		
			R	G	B	R	G	B
Horizontal	0.9275	0.0127	0.9892	0.9764	0.9726	−0.0159	−0.0052	−0.0255
Vertical	0.9627	−0.0190	0.9822	0.9694	0.9653	−0.0045	0.0385	−0.0537
Diagonal	0.9079	−0.0012	0.9736	0.9523	0.9508	−0.0032	0.0345	0.0216

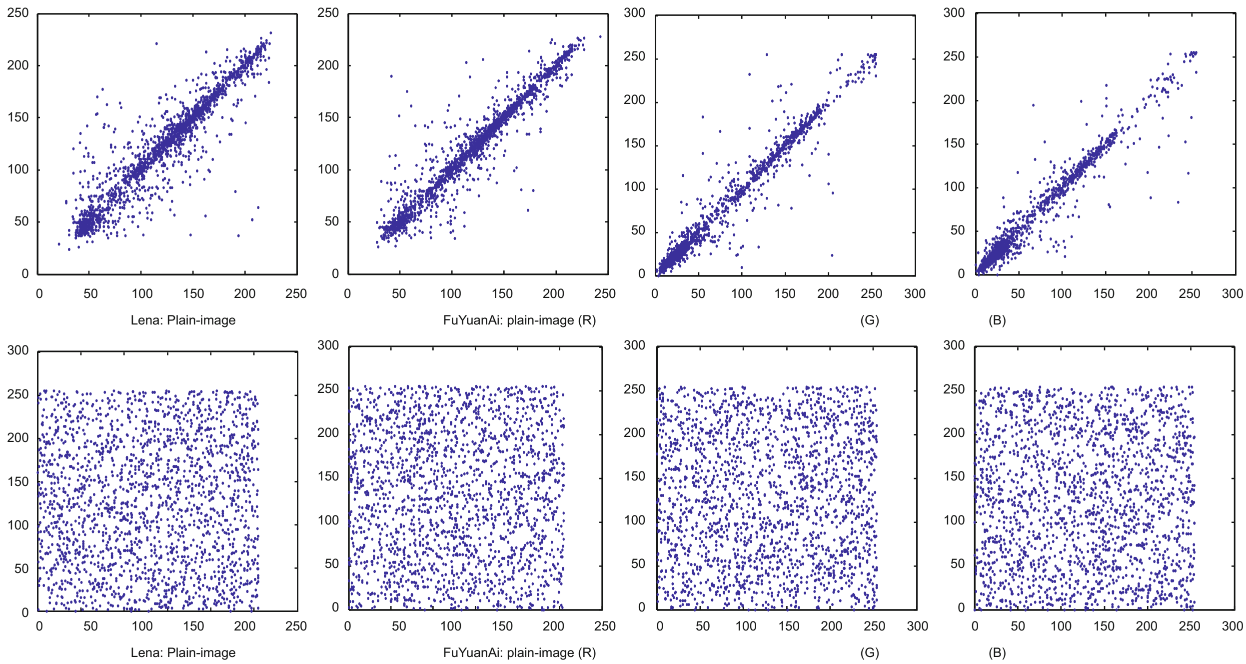

Fig. 7. Correlation of two horizontally adjacent pixels, x-coordinate and y-coordinate is the gray level of two neighbor pixels, respectively.

Table 2
The NPCR and UACI at different rounds.

Round	Lena	FuYuanAi		
		R	G	B
NPCR				
1	0.9979	0.9972	0.9966	0.9973
2	0.9965	0.9974	0.9967	0.9963
UACI				
1	0.3347	0.3320	0.3461	0.3331
2	0.3348	0.3332	0.3365	0.3325

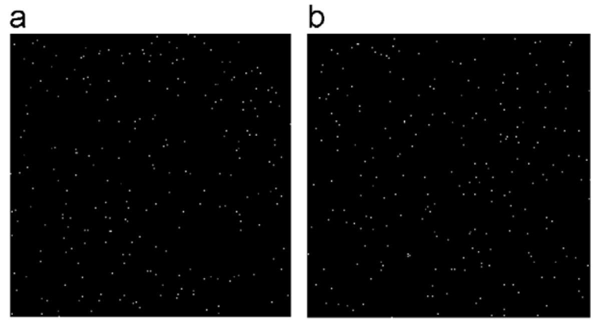

Fig. 8. Differences between two ciphered images. White points (about 1/256 of the total pixels) indicate the positions where pixels of the two cipher-images have the same values. (a) Lena, and (b) FuYuanAi—R plane.

85 to 84. Then the value of key sensibility at different rounds is calculated and listed in Table 4.

4.4. Encryption time

With the same performance requirement (NPCR > 0.995, UACI > 0.334), Table 5 compares the encryption time of the proposed scheme (2 rounds), Kwok-Wo Wong's scheme [8] (the value of its parameters: m=6, n=3) and Shiguo Lian's scheme [4] (the value of its parameters: m=1, n=4). The experiments are all performed using MATLAB 7.0 on a personal computer (PC) with a 1.8 GHz Celeronl processor, 0.99 GB memory and 80 GB harddisk capacity.

Table 3
Comparison of NPCR and UACI.

Round		Kwok-Wo Wong's scheme			Shiguo Lian's scheme			Proposed scheme
		$n=2$	$n=3$	$n=4$	$n=2$	$n=3$	$n=4$	
NPCR and UCAI								
1	NPCR	0.6866	0.9944	0.9960	0.0002	0.0003	0.0004	0.9979
	UACI	0.2088	0.3281	0.3349	0.0000	0.0001	0.0001	0.3347
2	NPCR	0.9961	0.9960	0.9960	0.0100	0.0198	0.0319	0.9965
	UACI	0.3343	0.3342	0.3349	0.0026	0.0051	0.0084	0.3348

Table 4
The key sensibility at different rounds.

Round	Lena	FuYuanAi		
		R	G	B
Sensibility of key_1 (1)				
1	0.9962	0.9957	0.9960	0.9958
2	0.9959	0.9961	0.9972	0.9966
Sensibility of key_2 (2)				
1	0.9966	0.9961	0.9959	0.9960
2	0.9960	0.9961	0.9967	0.9962
Sensibility of key_3 (3)				
1	0.9963	0.9964	0.9963	0.9956
2	0.9958	0.9963	0.9970	0.9990
Sensibility of key_4 (4)				
1	0.9964	0.9963	0.9961	0.9958
2	0.9959	0.9962	0.9969	0.9976

Table 5
The encryption time (unit: s).

Kwok-Wo Wong's scheme	Shiguo Lian's scheme	Proposed scheme
0.712	3.704	0.312

As the table shows, our algorithm is 2 times faster than Kwok-Wo Wong's Scheme, 10 times faster than Shiguo Lian's scheme. What's more, as far as parallel encryption are concerned, the time consumption will be much less.

5. Security and performance analysis

5.1. Diffusion

It can be seen that the values of NPCR are all more than 0.995 even in the first round. The NPCR analysis has revealed that, when there is only 1 bit of the plain pixel changed, almost all the cipher pixels become different although there is still a low possibility of 1/256 that two cipher pixels are equal. This diffusion first attributes to self adaptive encryption since change of any bit of the plain pixel will influences the amplitudes (A) of a wave as can be seen in the encryption processes. Then, the diffusion is spread out by wave transmission, which is the key process that contributes to diffusion.

5.2. Confusion

The histogram and correlation analyzes of adjacent pixels both indicate that our scheme possesses a good property of confusion. This mainly results from the pseudo-randomness of the key schedule and the wave transmission. They work together to introduce the random-like effect to the cipher image.

5.3. Encryption speed

Table 5 shows that our algorithm can achieve high speed. There are two mainly contributing factors:

(1) Only simple operations are included in our algorithm. In each round, for each pixel, there are only a few XOR operations, addition, etc. However, for the Shiguo Lian's scheme, there are three standard maps and one chaos iteration while for the Kwok-Wo Wong's scheme, there are four standard maps (at the same time do the value substitution) and one chaos iteration.
(2) Only 2-round encryption is enough to take on a perfect cipher-image with high security for our algorithm. However, for Shiguo Lian's scheme 6 rounds are needed. Though only one round are needed in Kwok-Wo Wong's scheme, there are much more operations within Kwok-Wo Wong's scheme than ours in one round.
(3) The algorithm can be executed in parallel.

Therefore, as far as the encryption speed is concerned, our scheme is very excellent.

5.4. Brute-force attack

For Brute-force attack, the security of our algorithm depends on the size of key place and the key sensibility.

Key space: As mentioned above, the total length of our key is 128 bits, which is by far very safe for ordinary business applications.

Key sensibility: According to Table 3, even in the first round, the key sensibility is more than 0.995 for each key, which means our algorithm is very sensible to the keys.

Therefore, our scheme is strong enough to resist brute-force attack.

5.5. *Known-plaintext attack*

As our algorithm is a kind of self-adaptive encryption, it is strong under known-plaintext attack [13]. So, our algorithm is more or at least the same secure compared with the Kwok-Wo Wong's scheme [8] and Shiguo Lian's scheme [4].

6. Conclusions

This paper presents a novel image encryption algorithm based on self-adaptive wave transmission, which can be easily realized and is computationally simple while achieving high security level, high speed, high sensitivity and other properties simultaneously. Wave transmission encryption, as the name suggests, is a way to change the gray-level value of pixels by simulating waves "transmitting" on the image. Self-adaptive is carried out by partition an image into two equal part, then use the information of one part to encrypt the other, respectively. In our algorithm, the information of one part is used to control the amplitude of the wave when encrypting the other part. Since the conventional CBC-like mode is eliminated, the proposed algorithm can encrypt image in parallel. Simulation results for gray-level image and color image demonstrate the high performance (sensitivity, speed, and security). We can see that, even in the first round the NPCR and key sensibility are already very high, only 2 rounds encryption can satisfy the performance and security requirement.

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